TRANSSHIPMENT MODELS

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Overview



The Transshipment Model

References

Table of Contents

1 The Transshipment Model



Introduction

1. The transshipment model is an extension of the transportation model in which intermediate transshipment points are added between the sources and destinations.
2. An example of a transshipment point is a distribution center or warehouse located between plants and stores.
3. In a transshipment problem, items may be transported from sources through transshipment points on to destinations, from one source to another, from one transshipment point to another, from one destination to another, or directly from sources to destinations, or some combination of these alternatives.

THE TRANSSHIPMENT MODEL



1. We will expand our wheat-shipping example to demonstrate the formulation of a transshipment model.

Example

1. Wheat is harvested at farms in Nebraska and Colorado before being shipped to the points. The amount of wheat harvested at each farm is 300 tons. The wheat is then shipped to the mills in Chicago, St. Louis, and Cincinnati. The shipping costs from the grain elevators to the mills remain the same, and the shipping costs from the farms to the grain elevators are as follows:

The Transshipment Model



Example (Cont.)

		Grain Elevator	
Farm	3. Kansas City	4. Omaha	5. Des Moines
1. Nebraska	\$16	\$10	\$12
2. Colorado	15	14	17

1. The basic structure of this model is shown in the figure below.

The Transshipment Model



Example (Cont.)

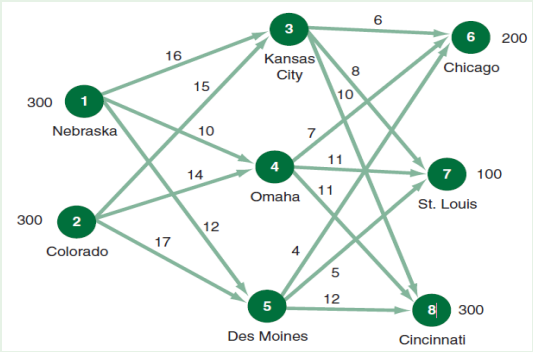


Figure: Network of transshipment routes.

The Transshipment Model



Example (Cont.)

1. As with the transportation problem, a linear programming model is developed with supply and demand constraints.
2. The available supply constraints for the farms in Nebraska and Colorado are
   - $x_{13} + x_{14} + x_{15} = 300$
   - $x_{23} + x_{24} + x_{25} = 300$



Example (Cont.)

1. The demand constraints at the Chicago, St. Louis, and Cincinnati mills are
   - $x_{36} + x_{46} + x_{56} = 200$
   - $x_{37} + x_{47} + x_{57} = 100$
   - $x_{38} + x_{48} + x_{58} = 300$

The Transshipment Model



Example (Cont.)

1. At each trans-shipment point, whatever is shipped in must also be shipped out.
2. Thus, the following constraints are developed for the trans-shipment points.
   - $x_{13} + x_{23} - x_{36} - x_{37} - x_{38} = 0$
   - $x_{14} + x_{24} - x_{46} - x_{47} - x_{48} = 0$
   - $x_{15} + x_{25} - x_{56} - x_{57} - x_{58} = 0$
3. The complete linear programming model, including the objective function, is summarized as follows:

The Transshipment Model

Example (End)

$$
\begin{aligned}
\texttt{Min.} \quad & Z = 16x_{13} + 10x_{14} + 12x_{15} + 15x_{23} + 14x_{24} \\
& +17x_{25} + 6x_{36} + 8x_{37} + 10x_{38} + 7x_{46} \\
\textbf{Objective}\uparrow \quad & +11x_{47} + 11x_{48} + 4x_{56} + 5x_{57} + 12x_{58} \\
\texttt{S.t.} \quad & \\
& x_{13} + x_{14} + x_{15} = 300 \\
\textbf{Supply}\uparrow \quad & x_{23} + x_{24} + x_{25} = 300 \\
& x_{36} + x_{46} + x_{56} = 200 \\
& x_{37} + x_{47} + x_{57} = 100 \\
\textbf{Demand}\uparrow \quad & x_{38} + x_{48} + x_{58} = 300 \\
& x_{13} + x_{23} - x_{36} - x_{37} - x_{38} = 0 \\
& x_{14} + x_{24} - x_{46} - x_{47} - x_{48} = 0 \\
\textbf{Transshipment}\uparrow \quad & x_{15} + x_{25} - x_{56} - x_{57} - x_{58} = 0 \\
\textbf{Non-negativity}\rightarrow \quad & x_{ij} \geq 0
\end{aligned}
$$

The Trans-shipment Solution process.



1. We can find an optimal solution to the transshipment problem by solving a transportation problem.
2. The steps are shown below:

Note the following:

1. Shipments to the dummy and from a point to itself will have a zero shipping cost.
2. Supply and demand at all transshipment points equal the total supplies at all sources.
3. Each supply point will have a supply equal to its original supply, and each demand point will have a demand equal to its original demand.
4. Balance the problem where necessary.

The Trans-shipment Solution process.



Example (Cont'd)

1. The following table shows the per unit cost between nodes:

	Destinations					
Sources	3	4	5	6	7	8
1	16	10	12	16	17	21
2	15	14	17	21	22	25
3	0	13	10	6	8	10
4	13	0	11	7	11	11
5	10	11	0	4	5	12

1. These values were obtained from the network.
2. $16, 17, 21, 21, 22$ and 25 are the minimum costs for X_{16}, X_{17}, X_{18}, X_{26}, X_{27} and X_{28}, respectively, choosing the best alternatives.
3. Below is the initial simplex tableau for the problem.



The Initial Transportation Tableau

11	To						
From	3	4	5	6	7	8	**Supply**
1	16	10	12	16	17	21	300
2	15	14	17	21	22	25	300
3	0	13	10	6	8	10	600
4	13	0	11	7	11	11	600
5	10	11	0	4	5	12	600
Demand	600	600	600	200	100	300	2400

Trans-shipment Solution process.



Solution cont'd

1. The next tableau shows the **least cost** solution to the problem.

The Least Cost Solution

12	To						
From	3	4	5	6	7	8	**Supply**
1	16	10	12	16 (200)	17 (100)	21	300
2	15	14	17	21	22	25 (300)	300
3	0 (600)	13	10	6	8	10	600
4	13	0 (600)	11	7	11	11	600
5	10	11	0 (600)	4	5	12	600
Demand	600	600	600	200	100	300	$12,400

Trans-shipment Solution process.



Solution cont'd

1. Since degeneracy is seen in **tableau** 12, we must make four empty cells basic variables with zero allocations.
2. This result is shown in **tableau** 13 from which we shall derive the **MODI** equations.

The Least Cost Solution

13	To						
From	3	4	5	6	7	8	**Supply**
1	16	10	12 (0)	16 (200)	17 (100)	21	300
2	15	14	17	21	22	25 (300)	300
3	0 (600)	13	10	6 (0)	8	10 (0)	600
4	13	0 (600)	11	7 (0)	11	11	600
5	10	11	0 (600)	4	5	12	600
Demand	600	600	600	200	100	300	$12,400

The Transshipment Solution Process



THE MODI

$$u_1 = 0 \quad (1)$$

$$u_1 + v_5 = 12 \quad (2)$$

$$u_1 + v_6 = 16 \quad (3)$$

$$u_1 + v_7 = 17 \quad (4)$$

$$u_2 + v_8 = 25 \quad (5)$$

EQUATIONS

$$u_3 + v_3 = 0 \quad (6)$$

$$u_3 + v_6 = 6 \quad (7)$$

$$u_3 + v_8 = 10 \quad (8)$$

$$u_4 + v_4 = 0 \quad (9)$$

$$u_4 + v_6 = 7 \quad (10)$$

$$u_5 + v_5 = 0 \quad (11)$$

The Transshipment Solution Process



SOLUTION

1. Putting Equation $(38) \rightarrow (39,\ 40,\ 41)$:

$$v_5 = 12,\ \ v_6 = 16 \tag{12}$$

$$v_7 = 17 \tag{13}$$

1. Putting Equation $(49,\ 50) \rightarrow (44,\ 47,\ 48)$:

$$u_3 = -10,\ \ u_4 = -9 \tag{14}$$

$$u_5 = -12 \tag{15}$$

END

1. Putting Equation $(51,\ 52) \rightarrow (43,\ 45,\ 46)$:

$$v_3 = 10,\ \ v_4 = 9 \tag{16}$$

$$v_8 = 20 \tag{17}$$

1. Putting Equation $(54) \rightarrow (42)$:

$$u_2 = 5 \tag{18}$$



COST CHANGE (k_{ij})

1. Given that

$$k_{ij} = c_{ij} - u_i - v_j \tag{19}$$

$$k_{13} = c_{13} - u_1 - v_3$$

$$k_{13} = 16 - 0 - 10 = \$\boxed{+6} \tag{20}$$

$$k_{14} = c_{14} - u_1 - v_4$$

$$k_{14} = 10 - 0 - 9 = \$\boxed{+1} \tag{21}$$

Cont'd

$$k_{23} = c_{23} - u_2 - v_3$$

$$k_{23} = 15 - 5 - 10 = \$\boxed{0} \tag{22}$$

$$k_{24} = c_{24} - u_2 - v_4$$

$$k_{24} = 14 - 5 - 9 = \$\boxed{0} \tag{23}$$

$$k_{25} = c_{25} - u_2 - v_5$$

$$k_{25} = 17 - 12 - 5 = \$\boxed{0} \tag{24}$$

The Transshipment Solution Process



COST CHANGE (k_{ij})

$$k_{26} = c_{26} - u_2 - v_6$$

$$k_{26} = 21 - 5 - 16 = \$\boxed{0} \quad (25)$$

$$k_{27} = c_{27} - u_2 - v_7$$

$$k_{27} = 22 - 5 - 17 = \$\boxed{0} \quad (26)$$

$$k_{34} = c_{34} - u_3 - v_4$$

$$k_{34} = 13 + 10 - 9 = \$\boxed{+14} \quad (27)$$

Cont'd

$$k_{35} = c_{35} - u_3 - v_5$$

$$k_{35} = 10 + 10 - 12 = \$\boxed{+8} \quad (28)$$

$$k_{37} = c_{37} - u_3 - v_7$$

$$k_{37} = 8 + 10 - 17 = \$\boxed{+1} \quad (29)$$

$$k_{43} = c_{43} - u_4 - v_3$$

$$k_{43} = 13 + 9 - 10 = \$\boxed{+12} \quad (30)$$

The Transshipment Solution Process

COST CHANGE (k_{ij})

$$k_{45} = c_{45} - u_4 - v_5$$

$$k_{45} = 11 + 9 - 12 = \$\boxed{+8} \quad (31)$$

$$k_{47} = c_{47} - u_4 - v_7$$

$$k_{47} = 11 + 9 - 17 = \$\boxed{+3} \quad (32)$$

$$k_{48} = c_{48} - u_4 - v_8$$

$$k_{48} = 11 + 9 - 20 = \$\boxed{0} \quad (33)$$

Cont'd

$$k_{53} = c_{53} - u_5 - v_3$$

$$k_{53} = 10 + 12 - 10 = \$\boxed{+12} \quad (34)$$

$$k_{54} = c_{54} - u_5 - v_4$$

$$k_{54} = 11 + 12 - 9 = \$\boxed{+14} \quad (35)$$

$$k_{56} = c_{56} - u_5 - v_6$$

$$k_{56} = 4 + 12 - 16 = \$\boxed{0} \quad (36)$$

The Transshipment Solution Process



COST CHANGE (END)

$$k_{57} = c_{57} - u_5 - v_7$$

$$k_{57} = 5 + 12 - 17 = \$\boxed{0} \quad (37)$$

$$k_{58} = c_{58} - u_5 - v_8$$

$$k_{58} = 12 + 12 - 20 = \$\boxed{+4} \quad (38)$$

1. The current solution is optimal since all cost changes $k_{ij} \geq 0$.

Remarks:

1. Due to the degenerate condition in tableau $\boxed{12}$, more iterations may be done when tracing the optimal solution.
2. This depends on the choice of arbitrary basic variables from the degenerate tableau.
3. See the shipping list and optimal transportation network below.

The transshipment Solution Process

SHIPPING LIST

Source	Destination	Shipments	Shipping Cost
1	6	200	$200 \times 16 = \$3,200$
1	7	100	$100 \times 17 = \$1,700$
1	5	0	$0 \times 12 = 0$
2	8	300	$300 \times 25 = \$7,500$
3	3	600	$600 \times 0 = 0$
3	6	0	$0 \times 6 = 0$
3	8	0	$0 \times 10 = 0$
4	4	600	$600 \times 0 = 0$
4	6	0	$0 \times 7 = 0$
5	5	600	$600 \times 0 = 0$
Total Cost			**$12,400**



The Network Solution

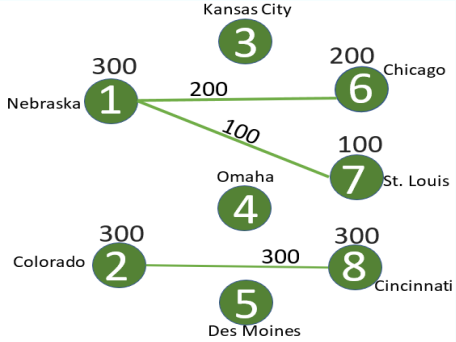


Figure: The Transshipment Network Solution.



Exercises

Transshipment Solution Process



Exercise

1. Consider tableau 13 and use the **Stepping Stone method** to determine the optimal solution to the trans-shipment problem above.
2. Determine at least one alternate solution (excluding the one on slide 76).

Bibliography



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